Literature Review

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1 Introduction

After discussion with A/Prof Douglas R. Brumley, My literature review will be around Ciliary Flow.

2 Ciliary Flow

Definition 2.1 (Cilium (pl. Cilia))**.** In the context of eukaryotes, *cilia* are short, hair-like protrusions that grow from the surface as an extension of the membrane.

Cilia serve important functions in motility, sensory reception, signalling, and mass transport. Studies of ciliary flows therefore have significant biological implications. [1]

2.1 Corals

Cilia-induced vortical flows are critical for O_2 regulation and metabolite exchange across coral-water interfaces. Pacherres et al. [2]

- Methodology: "high-speed imaging with a microscope chamber to determine cilia beating frequencies (CBFs) and particle image velocimetry (PIV) in combination with chemical imaging (SensPIV) for mapping flow and O2 concentration fields of corals exposed to increasing seawater temperature."
- Found that ciliary beating is important for thermal tolerance for corals. By beating with higher frequency, the corals increase O_2 intake, which alleviates hypoxia observed in temperatures ~ 35 C° and above. Above ~ 37 C°, coral morality accelerated, indicating the existence of a physiological tipping-point for reef-building corals.

A shortcoming of this study and its conclusions is that it is unclear what specific temperature-dependent dynamic viscosity function was used in the Navier-Stokes equation for the mathematical simulations. Furthermore, as highlighted in a study of mussels by Riisgård and Larsen [3], higher temperatures lead to low viscosity, which in term lead to higher cilia beating frequency.

2.2 Synchronisation

References

[1] Naoko Mizuno et al. "Structural Studies of Ciliary Components". In: *Journal of Molecular Biology* 422.2 (Sept. 2012), pp. 163–180. ISSN: 0022-2836. DOI: 10.1016/j.jmb.2012.05.040.

[2] Cesar O. Pacherres et al. "Acute Temperature Effects on Cilia Beating Increase Coral Deoxygenation". In: *Science Advances* 12.21 (), eaeg0950. DOI: 10.1126/sciadv.aeg0950. (Visited on 09/02/2026).

[3] Hu Riisgård and Ps Larsen. "Viscosity of Seawater Controls Beat Frequency of Water-Pumping Cilia and Filtration Rate of Mussels Mytilus Edulis". In: *Marine Ecology Progress Series* 343 (Aug. 2007), pp. 141–150. ISSN: 0171-8630, 1616-1599. DOI: 10.3354/meps06930. (Visited on 08/28/2026).